ENHANCING THREAT DETECTION AND RESPONSE IN HEALTHCARE CYBERSECURITY

AN INVESTIGATIVE MALWARE ANALYSIS

CONDUCTED BY

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FOR

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Executive Summary

This report presents the outcome of a targeted threat-intelligence investigation initiated in response to a suspected malware campaign affecting healthcare organisations. Using automated hash analysis via the VirusTotal API, one malicious file hash was conclusively identified and attributed to the MedusaLocker ransomware family.

Technical analysis determined that MedusaLocker is a high-impact ransomware strain known for enterprise targeting, data encryption, lateral movement, credential compromise, and double-extortion operations. The malware has demonstrated sustained activity across healthcare and other critical-infrastructure sectors, posing significant operational, financial, and regulatory risk.

A comprehensive intelligence-gathering effort was conducted to identify associated Indicators of Compromise (IoCs), including malicious IP infrastructure, attacker communication emails, cryptocurrency wallets, ransomware artefacts, and TOR negotiation portals. All indicators were validated across multiple threat-intelligence platforms to ensure analytical confidence.

To strengthen proactive detection, a Suricata rule was developed and successfully tested to identify outbound communication with known malicious infrastructure. The findings of this investigation provide actionable intelligence, enhanced detection capability, and strategic security recommendations to improve organisational resilience against ransomware threats.

Introduction

SecureHealth Cyber Solutions, a leading cybersecurity firm that specialises in providing high-end threat intelligence services to healthcare organisations has been alerted to a potential malware campaign targeting healthcare organisations. These cyberattacks are believed to be orchestrated by a sophisticated threat group with a history of exploiting vulnerabilities in healthcare IT infrastructure.

The Security Operations Center (SOC) has intercepted a collection of suspicious file hashes suspected to be linked to recent cyberattacks. However, intelligence reports indicate that only one of these files is malicious, but its details remain unknown.

As part of the Threat Intelligence team, my mission is to analyse these file hashes, identify the malicious one, and investigate its origins and associated Indicators of Compromise (IoCs). My findings will be presented to the executive leadership, emphasising the impact of the threat and how to defend against it.

Scope

This investigative analysis will cover the following:

1. Analysis of suspicious file hashes to identify the malicious file.
2. Further research on identified malicious file.
3. Research Indicators of Compromise such as IPs, email addresses, cryptocurrency wallets, file extensions, ransom notes, and TOR addresses associated with the malicious file.
4. Implementation of IDS (Suricata) to detect malicious IP triggered by hping3 (simulated network traffic).
5. Findings and recommendations.

Tools and Environment

Kali Linux and Ubuntu VM, OSINT platforms such as VirusTotal, Suricata IDS version 8.0.3, hping3 for traffic simulation, VirusTotal API key, cURL and jq for API automation.

Methodology

Phase 1: Malware Identification

1. Create a bash script that contains Virus Total API key and suspicious file hashes.
2. Make script executable and run script in Kali VM to analyse file hashes.
3. Research identified malware including history and associated cyberattacks.

Phase 2: Threat Intelligence

1. Using OSINT techniques, research Indicators of Compromise (IOC) such as IP addresses, Email addresses, cryptocurrency wallet addresses, file extensions, ransom note file names, and TOR addresses linked to the identified malware.

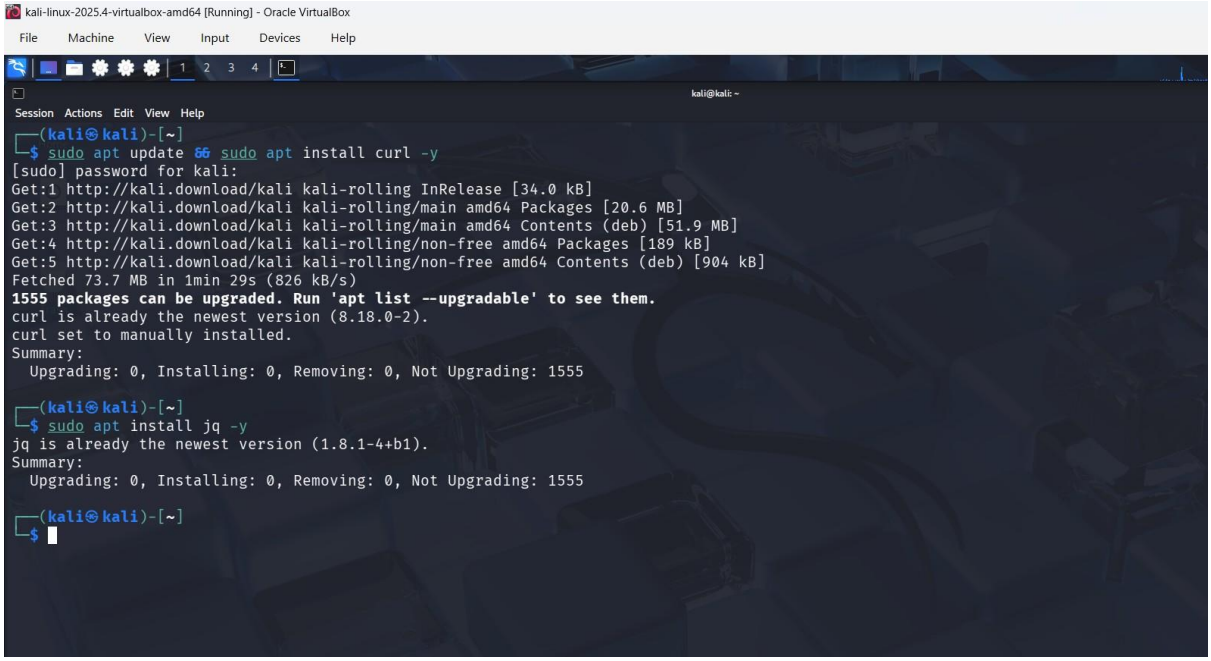
Phase 3: Suricata Rule Creation and Testing

1. Create a Suricata rule in Kali/Ubuntu VM using one of the IP addresses linked to the malicious file.
2. Test the rule in Suricata by simulating a network traffic using hping3 command created with the same malicious IP address.
3. Check Suricata logs to confirm alert was triggered successfully.

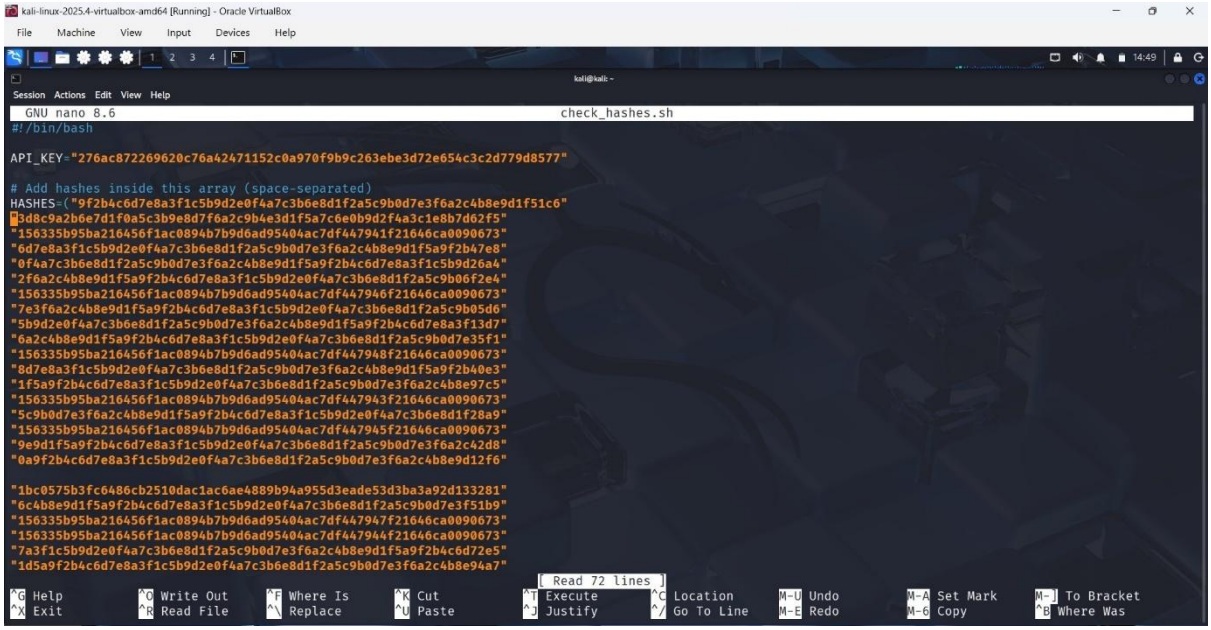
Phase 4: Comprehensive Report

Phase 1

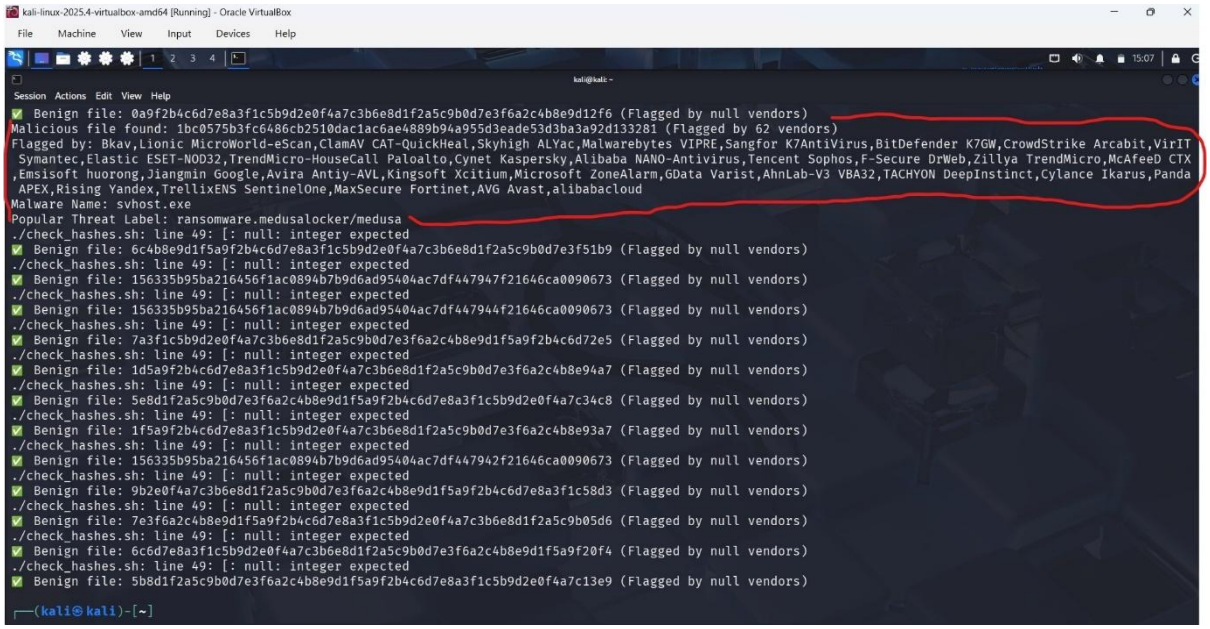
Initial configurations: cURL and jq



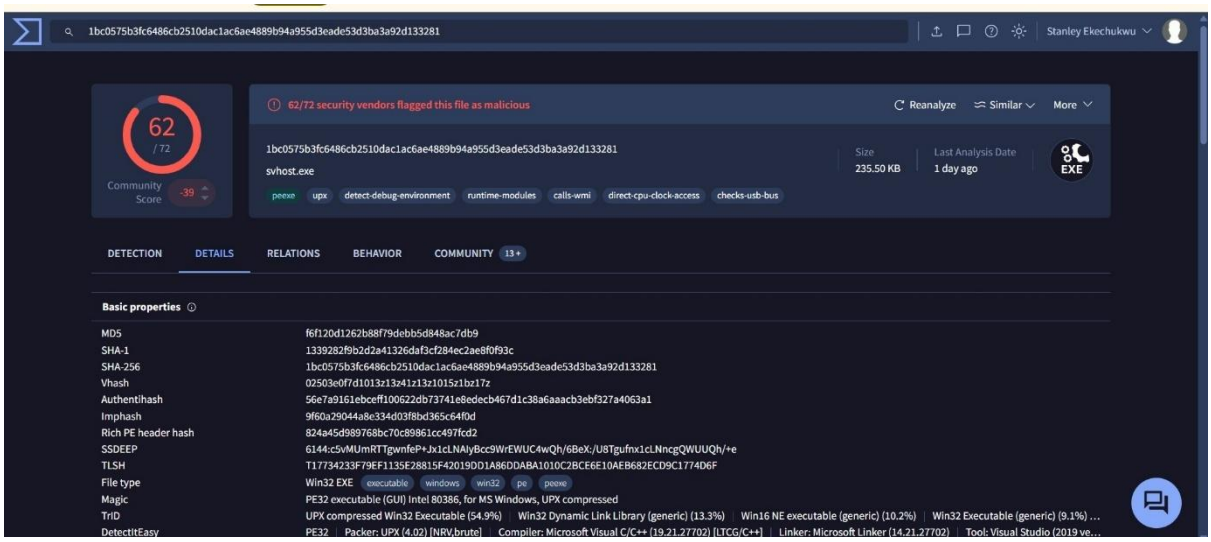
Suspicious file hashes ready to be analysed



Malicious file identified after analysis:
1bc0575b3fc6486cb2510dac1ac6ae4889b94a955d3eade53d3ba3a92d133281



Malware Name: svhost.exe
Flagged by 62 anti-virus vendors
Popular Threat Label: ransomware.medusalocker/medusa



Malware Investigation

This malicious file hash, also identified as SHA-256 is a ransomware binary. This exact SHA-256 is linked to **MedusaLocker ransomware,** and also shows a behaviour consistent with file encryption and extortion, which can be tied to typical properties of ransomware in general.

The MedusaLocker ransomware generally encrypts files on infected machines, displays a ransom note claiming the network has been attacked, and demands payment for decryption. This ransomware mostly uses RSA + AES encryption to prevent other options of recovery except the attacker's key. This particular attack and ransomware in general are usually

delivered through **phishing attachments, exploit kits, malicious macro documents, and compromised RDP credentials**

Once initial access is successful, the malware encrypts files and disrupts normal access, drops ransom demand files, and attempts to persist. It further proceeds to disable backups or shadow copies.

Cyberattacks linked to this malicious file hash are:

1. **Healthcare Sector :** This malicious file hash(MedusaLocker) has constantly targeted hospitals and medical facilities, especially during the COVID-19 period. **Hospitals in the United States** were forced to divert patients due to encrypted systems, **European healthcare providers** experiencing full network outages, and **medical imaging and diagnostic centres.**

2. **Critical Infrastructure and Municipal Governments:** This ransomware has also attacked city councils, local government offices, public service departments, and water/energy organisations.

3. **Manufacturing and Industrial Companies:** Manufacturers have experiences locked industrial control systems, production line shutdowns, encrypted design files and ERP systems due to this MedusaLocker ransomware.

4. **Education Sector:** The impact of this malicious file hash in colleges and universities have recorded encrypted student database, locked research databases and data leaks.

5. **Finance Sector:** Various institutions providing financial services, including loans, insurance and mortgages have experienced and are still experiencing the negative impact of MedusaLocker ransomware attacks, which includes encryption of financial records, theft of customer data, and threats to expose sensitive information on the dark web.

HISTORY AND EVOLUTION OF MEDUSALOCKER RANSOMWARE

The MedusaLocker ransomware was first observed in **September 2019**, with attributes such as **basic AES + RSA encryption, simple ransom notes, manual deployment by attackers, and heavy reliance on RDP compromise**

In 2020, MedusaLocker evolved significantly, with attributes such as sophisticated encryptions, improved persistence mechanisms, automated lateral movement scripts, better obfuscation, and communication with the dark web.

Around mid-late 2020, MedusaLocker adopted the double-extortion model where they began to steal data before encryption, threaten to leak sensitive data, and launched a TOR-based leak site, thereby creating multiple channels of extorting from their victims. This shift aligned MedusaLocker with major players like Maze, Revil, and NetWalker.

In 2021, MedusaLocker began showing signs of affiliate-driven intrusions, shared infrastructure, consistent payload but varied intrusion methods**(RaaS). Between 2021 and 2022**, operators of MedusaLocker ransomware increased their targets on healthcare and critical infrastructures such as hospitals, medical research centres, municipal governments, and industrial companies.

Between 2022 and 2024, threat intelligence records revealed that this ransomware continued to update codes resulting in **new file extensions, updated ransom notes, better evasion techniques, more robust encryption, and faster lateral movement across networks, privilege escalation, shadow copy deletion, UAC bypass techniques, and network share enumeration.**

Since 2024 till date (2026), MedusaLocker remains dangerous and
- Continues to target enterprises globally.
- Uses credential theft, RDP compromise, and VPN exploitation.
- Operates with a stable affiliate model.
- Maintains a functioning leak site.
- Frequently updates payloads to evade detection.

Phase 2: Threat Intelligence Collection

IOCs associated with the malware

IP Addresses:

135.232.92.137- Contacted during sandbox execution of the malicious file. It represents network activity related to the malware sample; may indicate attempted outbound connections for payload retrieval or C2 communication. While observed in VirusTotal relations, there is no confirmed OSINT linking it as a MedusaLocker-controlled server. Treat as a related IOC for forensic correlation, not definitive infrastructure.

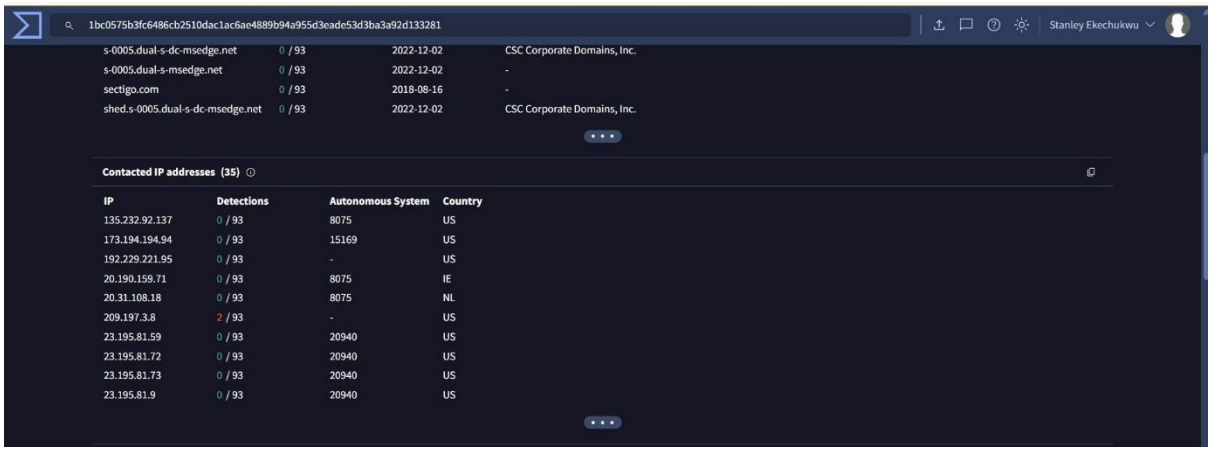
173.194.194.94– This appears in dynamic analysis as a contacted IP which could reflect benign network traffic or interaction with shared hosting/cloud infrastructure during malware execution. No public evidence links this IP directly to MedusaLocker ransomware; useful as a sandbox-observed IOC rather than confirmed malicious infrastructure.

192.229.221.95- Contacted by the malware in dynamic execution and recorded in VirusTotal relations. It may indicate secondary payload retrieval or staging activity during sample execution. Serves as a related IOC for forensic purposes, but attribution to MedusaLocker operators is unverified.

20.190.159.71- Outbound connection observed in sandbox/network analysis which could serve as a relay, proxy, or potential C2 communication endpoint. No public OSINT confirms MedusaLocker association; include as a related network IOC for investigation but not as confirmed infrastructure.

20.31.108.18- Logged in sandbox environments as contacted by the malicious file and could indicate secondary stage communications or beacon attempts from the malware. Treat as an associated IOC for monitoring or network correlation, not definitive ransomware C2 or staging host.

IP addresses linked to the malicious file as shown in Virus Total



Email Addresses:

sambolero@tutanota.com - Used in multiple MedusaLocker ransom notes and victims are instructed to contact this address to receive decryption instructions. It is hosted on **Tutanota**, providing anonymity to operators. Observed in malware analysis reports and threat intel feeds (enigmasoftware.com).

folielol@protonmail.com - One of the most commonly reported MedusaLocker contact emails. It appears in ransom notes during campaigns targeting various organisations. ProtonMail is used for encrypted, anonymous communication and confirmed in malware sample analysis and open OSINT repositories (salvagedata.com).

ctorsenoria@tutanota.com - Alternate contact email seen in ransom notes of MedusaLocker infections which enables operators to provide decryption guidance and manage ransom negotiations. Tutanota ensures secure and private communication and cited in malware IOC analyses for MedusaLocker (geeksadvice.com).

rightcheck@cock.li - Used in later MedusaLocker campaigns when other email providers were unavailable. It appears in ransom notes instructing victims on payment and decryption. Cock.li is a known "bulletproof" email provider, often abused by ransomware groups. Documented in malware sample examinations (enigmasoftware.com).

suppdecrypt@protonmail.com - Occasionally appears in MedusaLocker-related ransom notes and likely used for victim communication and proof-of-life messages. ProtonMail provides encrypted communication for operators. This is not officially listed in all IOC sources but observed in some community reports (medium.com).

These addresses have been observed within ransom note artefacts and victim communications across incident response reporting and sandbox analyses. Attribution is assessed at the campaign level rather than being exclusive to a single malware sample.

Cryptocurrency wallet addresses:

1AbRxRfP6yHePpi7jmDZkS4Mfpm1ZiatH5 – This is listed as a MedusaLocker ransom payment address in the joint FBI/CISA advisory on MedusaLocker ransomware IOCs. It is observed as one of the wallets used to receive ransom payments in tracked MedusaLocker attacks. It is used to help analysts cluster ransom payments and tie multiple incidents to the same ransomware family.

18wRbb94CjyTGkUp32ZM7krCYCB9MXUq42 – This appears in the official MedusaLocker IOC list published by CISA/FBI as a known Bitcoin payment wallet used by the group. It helps ransomware researchers map out payment flows and assess victimization patterns. This address is often included in blocklists used by defenders to monitor suspicious outgoing BTC transfers.

1DRxUFhvJjGUdojCzMWSLmwx7Qxn79XbJq – This wallet address is documented in the MedusaLocker ransomware ransomware advisory. Funds received here have been linked by analysts to ransomware demands following MedusaLocker infections. This address is used as part of tracking campaigns and correlating blockchain payments with incidents.

1Edcufenw1BB4ni9UadJpQh9LVx9JGtKpP – This is officially published in the FBI/CISA MedusaLocker IOC list as a malicious payment address. Records indicate that ransomware proceeds have been received at this address on several occasions. This is useful for threat hunters and incident responders when analyzing ransom payment evidence.

1DyMbw6R9PbJqfUSDcK5729xQ57yJrE8BC - Included in the MedusaLocker wallet list from the same CISA/FBI advisory. Represents one of the BTC addresses that researchers have observed in actual MedusaLocker ransom payment activity. It helps cybersecurity professionals correlate suspicious BTC flows with ransomware events.

File extensions used by the malware:

.encrypted – One of the *primary and earliest* file extensions used by MedusaLocker to mark encrypted files. It is appended to *all encrypted files* in many variants, indicating successful encryption by the ransomware. Technical analyses and malware reports confirm that .encrypted is a standard extension appended to files encrypted by MedusaLocker. It is used across multiple MedusaLocker samples as a signature indicator; it is widely used in forensic detection and automated threat hunting.

.bomber – A known variant extension used by MedusaLocker in certain campaigns. It *distinctively marks encrypted files* and appears in technical IOC lists from malware analysts. Multiple sources list .bomber as one of the extensions appended to encrypted files by MedusaLocker. This extension is unusual and specific which labels it as a strong IOC when found on victim files.

.breakingbad – This is another file extension appended by MedusaLocker during encryption. The name is memorable and has been documented in variants of this ransomware family. Threat intelligence summaries and ransomware IOCs include .breakingbad as part of the set of extensions used by MedusaLocker over time. It is a reliable indicator of MedusaLocker activity when seen on encrypted data.

.locker16 – One of the number-based extension variants seen in MedusaLocker samples. The numeric part (16) varies by campaign/version. Analyst reports list .locker16 among the extensions that MedusaLocker can append. Patterns like .lockerXX are associated with ransomware behaviour and are used in behavioural detection rulesets.

.skynet – Another extension used by MedusaLocker. It is recognised as part of the group of extensions that the ransomware may append post-encryption. Multiple technical sources include .skynet as part of extensions observed in MedusaLocker infections. Useful as an IOC in detection and incident response, especially when correlated with other ransomware behaviours.

Ransom Note File Names:

HOW_TO_RECOVER_DATA.html - One of the *most commonly observed* MedusaLocker ransom note filenames dropped in affected directories after encryption. Typically contains the ransom demand, contact information (email, TOR), and instructions for victims. It is cited in multiple technical analyses of MedusaLocker behaviour.

HOW_TO_OPEN_FILES.html - Documented in IOC lists from threat intelligence reporting as a MedusaLocker ransom note name. It serves the same purpose as other HTML notes: instructions for ransom payment and contacting attackers. It appears consistently across variants and sample captures.

READ_INSTRUCTION.html - Observed as a ransom note filename for MedusaLocker in public IOC lists. It reflects common wording used by MedusaLocker families to convey decryption instructions. Multiple case samples use this naming pattern.

INSTRUCTIONS.html - Generic but *confirmed* in multiple MedusaLocker variants as a ransom note name. It appears in IOC compilations alongside other HTML ransom notes and useful as an indicator when correlated with encrypted extensions and other behaviour.

READ_NOTE.html - Used in some documented MedusaLocker variants and related strains (such as "Stolen") in public analysis write-ups. It is equivalent in function to other HTML notes: explains victim situation and contact/payment instructions. Listed in ransomware guides and IOC collections.

TOR Addresses:

qd7pcafncosqfqu3ha6fcx4h6sr7tzwagzpcdcnytiw3b6varaeqv5yd.onion - This is the most widely seen *MedusaLocker* TOR negotiation/payment site used in current campaigns. It is referenced in malware analysis as the primary onion address victims are directed to after files are encrypted. It is a victim negotiation portal that hosts ransom instructions and contact channels**.**

z6wkgghtoawog5noty5nxulmmt2zs7c3yvwr22v4czbffdoly2kl4uad.onion - Reported as a *data leak / extortion demonstration site* for MedusaLocker. After victims refuse to pay, attackers may post stolen data to pressure payment. It shows proof of stolen data to coerce victims.

gvlay6u4g53rxdi5.onion -This older TOR address appears in official U.S. government ransomware IOC listings tied to MedusaLocker campaigns before the switch to newer v3 onion domains. It used to be a negotiation/payment portal before migration to TOR v3.

medusacegu2ufmc3kx2kkqicrlcxdettsjcenhjena6uannk5f4ffuyd.onion - Documented as another MedusaLocker-related negotiation or extortion site distinct from the primary "qd7pc…" address. This domain follows the same naming conventions used by the group and has been observed in some reporting.

medusaxko7jxtrojdkxo66j7ck4q5tgktf7uqsqyfry4ebnxlcbkccyd.onion - Seen alongside the previous domain as part of the MedusaLocker network infrastructure. It has been observed in dark-web references linked to MedusaLocker extortion operations. This is a secondary onion portal used by MedusaLocker for victim interaction.

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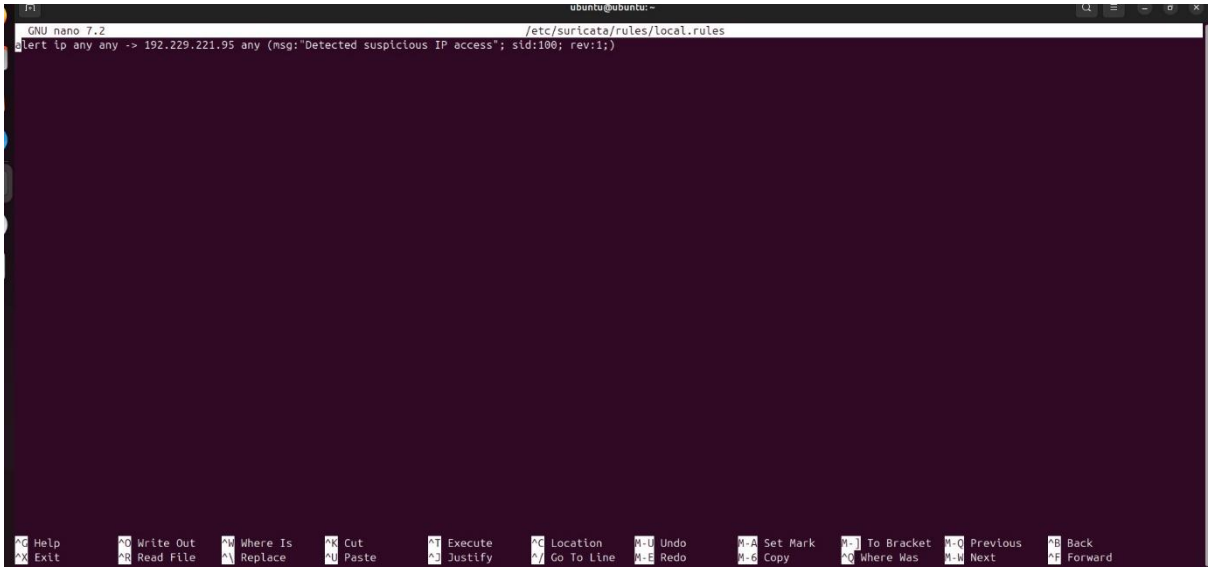
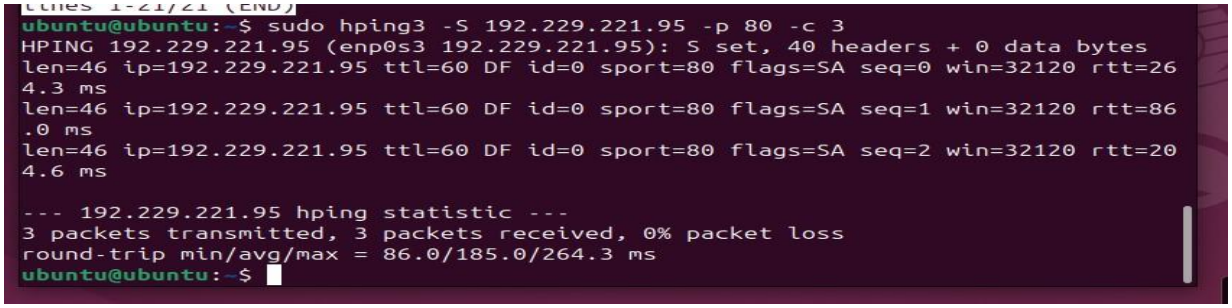
Phase 3: Suricata Rule Creation and Testing

Suricata rule using a malicious IP address associated with the malicious file hash
alert ip any any -> 192.229.221.95 any (msg:"Detected suspicious IP access"; sid:100; rev:1;)

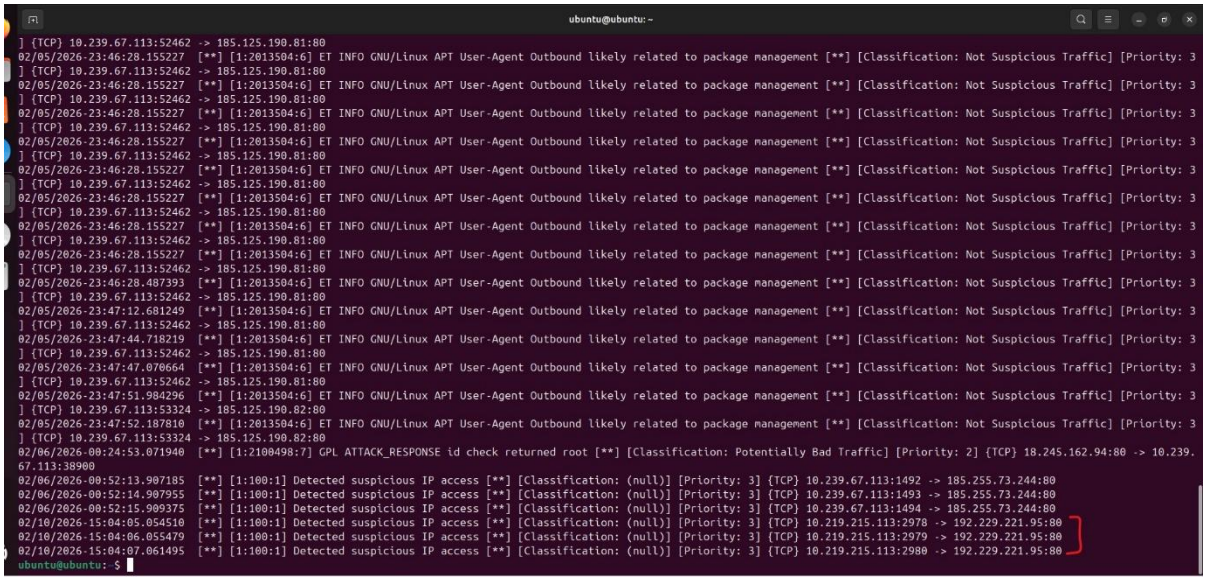


Network simulation using hping3

sudo hping3 -S 192.229.221.95 -p 80 -c 3



Suricata logs showing malicious IP linked to identified malicious file hash



```
ubuntu@ubuntu: ~
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:46:28.155227  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:46:28.155227  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
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02/05/2026-23:46:28.155227  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
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02/05/2026-23:46:28.155227  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:46:28.155227  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:46:28.487393  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:47:12.681249  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:47:44.718219  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:47:47.070664  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:52462 -> 185.125.190.81:80
02/05/2026-23:47:51.984296  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:53324 -> 185.125.190.82:80
02/05/2026-23:47:52.187810  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3
] {TCP} 10.239.67.113:53324 -> 185.125.190.82:80
02/06/2026-00:24:53.071940  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 18.245.162.94:80 -> 10.239.
67.113:38900
02/06/2026-00:52:13.907185  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.239.67.113:1492 -> 185.255.73.244:80
02/06/2026-00:52:14.907955  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.239.67.113:1493 -> 185.255.73.244:80
02/06/2026-00:52:15.909375  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.239.67.113:1494 -> 185.255.73.244:80
02/10/2026-15:04:05.054510  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.219.215.113:2978 -> 192.229.221.95:80
02/10/2026-15:04:06.055479  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.219.215.113:2979 -> 192.229.221.95:80
02/10/2026-15:04:07.061495  [**] [1:100:1] Detected suspicious IP access [**] [Classification: (null)] [Priority: 3] {TCP} 10.219.215.113:2980 -> 192.229.221.95:80
ubuntu@ubuntu:~$
```

Recommendations

1. Strengthen Email Security Controls
Deploy advanced phishing detection, attachment sandboxing, and malicious URL analysis. Enforce DMARC, DKIM, and SPF policies to mitigate spoofing and email-borne malware delivery.

2. Harden Remote Access Infrastructure
Disable unused RDP services, enforce Multi-Factor Authentication (MFA) across VPN and remote access platforms, and implement geo-restriction policies. Monitor continuously for brute-force and credential-stuffing activity.

3. Enhance Endpoint Detection and Response
Deploy enterprise-grade EDR/XDR solutions capable of detecting ransomware behaviours such as mass encryption, shadow-copy deletion, and privilege escalation. Enforce application allow-listing and restrict unsigned binaries.

4. Expand Network Detection Coverage
Ingest all identified IoCs into SIEM and SOAR platforms. Deploy Suricata signatures across network sensors and monitor for TOR traffic, anomalous outbound connections, and beaconing patterns.

5. Implement Robust Backup and Recovery Strategy
Maintain offline and immutable backups, restrict administrative access to backup repositories, and routinely test restoration procedures to ensure operational continuity.

6. Strengthen Vulnerability and Patch Management
Prioritise remediation of externally exposed services, including VPN gateways and RDP hosts. Conduct recurring vulnerability assessments and track remediation timelines.

7. Improve Security Awareness and Training
Conduct phishing simulations and ransomware awareness programmes to improve user vigilance and reporting speed.

8. Advance Incident Response Preparedness
Update ransomware response playbooks, establish rapid isolation procedures, and maintain coordination readiness with legal counsel, cyber-insurance providers, and law enforcement agencies.

Conclusion

This investigation successfully identified and attributed a malicious file hash to the MedusaLocker ransomware family, confirming the presence of a sophisticated threat capable of disrupting healthcare operations and compromising sensitive patient data. Through structured threat-intelligence analysis, critical insight was gained into the malware's behaviour, infrastructure, and extortion methodology.

The identification and validation of multi-layered Indicators of Compromise enabled the development of actionable detection logic, including a tested Suricata rule designed to detect communication with malicious infrastructure. These detection measures enhance the organisation's ability to identify and contain ransomware activity at earlier stages of the attack lifecycle.

Given MedusaLocker's continued evolution and its focus on high-value sectors such as healthcare, sustained defensive vigilance is required. Implementing the strategic and technical recommendations outlined in this report will significantly strengthen security posture, improve detection maturity, and support rapid incident containment.

Proactive threat intelligence, layered security controls, and resilient recovery capabilities remain essential to safeguarding patient data, maintaining operational continuity, and ensuring regulatory compliance amid persistent ransomware threats.